GUEST LECTURE · DESIGN THINKING

The Role of Generative AI in the Design Thinking Process

What happens when human-centered innovation meets generative AI

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About Me & The NOVA Innovation and Analytics Lab

Who I am

I hold an MSc in Data-Driven Marketing from NOVA Information Management School, where my thesis explored the role of generative AI within the Design Thinking process — supervised by Professor Guilherme Victorino and Professor João Valente Cordeiro. My work sits at the intersection of human-centered design, data and emerging AI tools, and today's session brings that research directly to you.

The NOVA Innovation and Analytics Lab

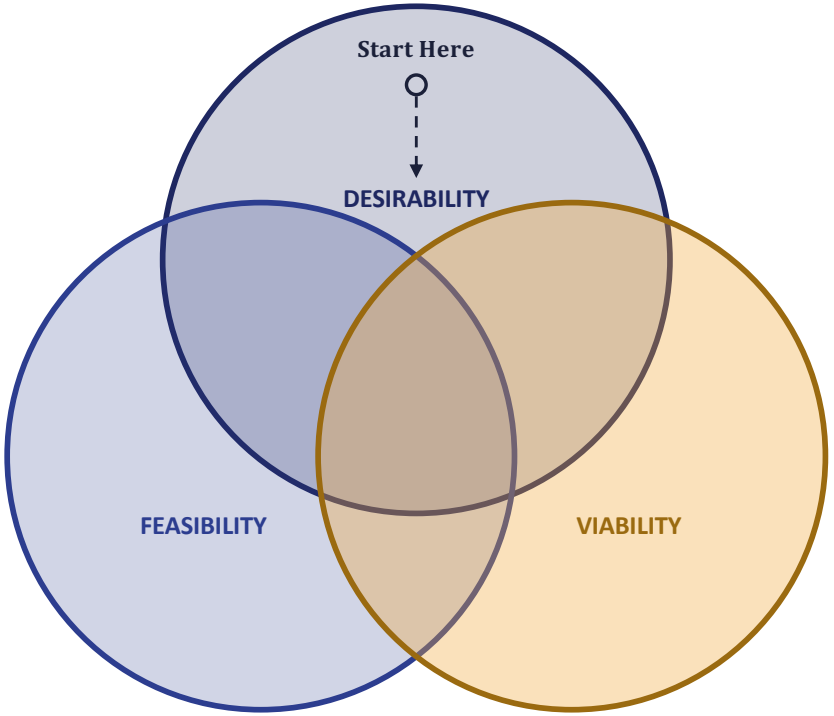
The Lab where this research was developed and tested. It works at the intersection of people and data — combining Data Science, Design Thinking and Behavioral Sciences to turn new insights into disruptive ideas and scalable solutions, for NOVA's community and its partners. It champions empathy, creativity, visualization and storytelling as tools for better decision-making.

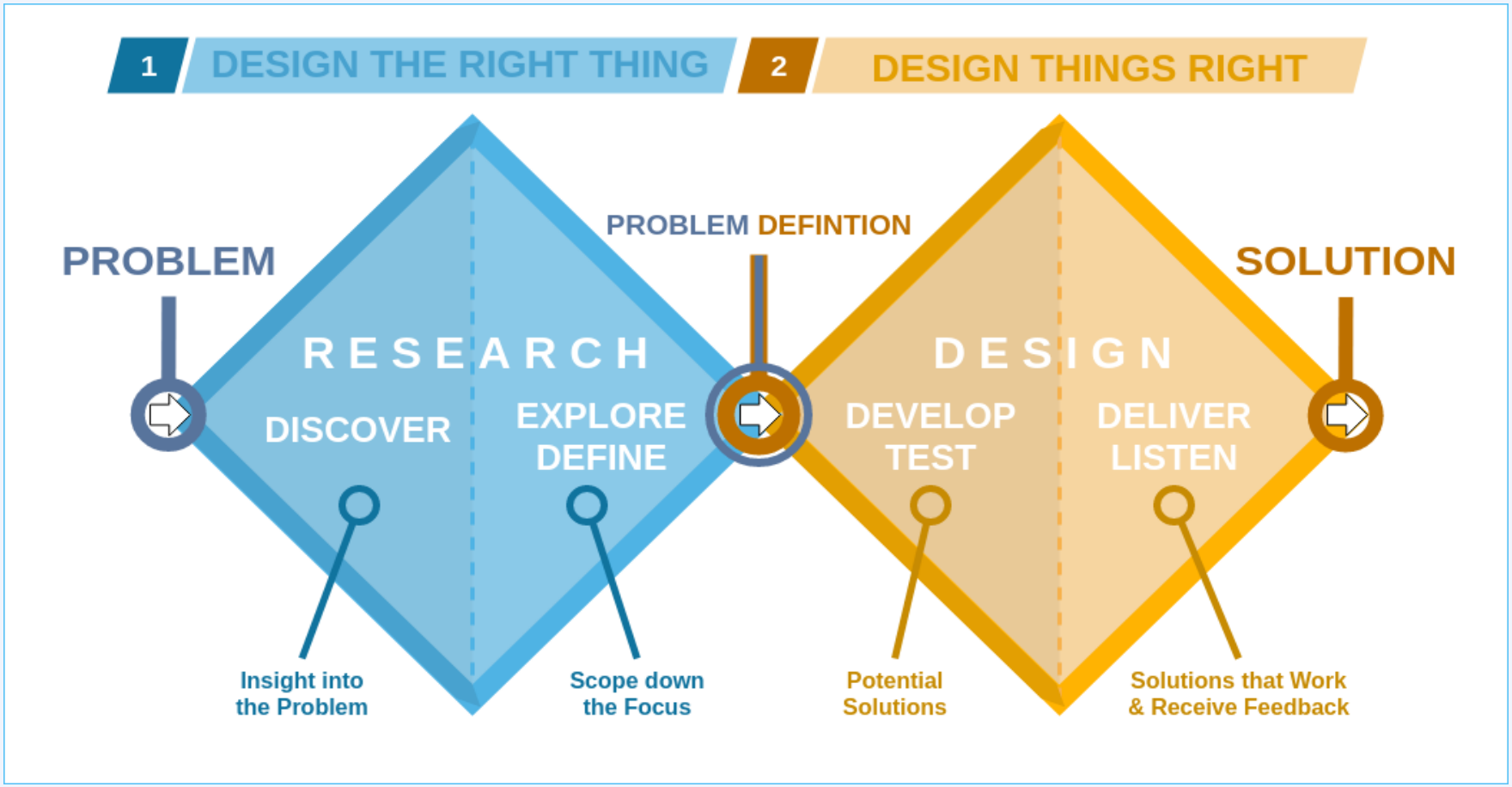
Translate Ideas Into Action

During the early phases of the process, we focus on experiments that test the **desirability** of the idea.

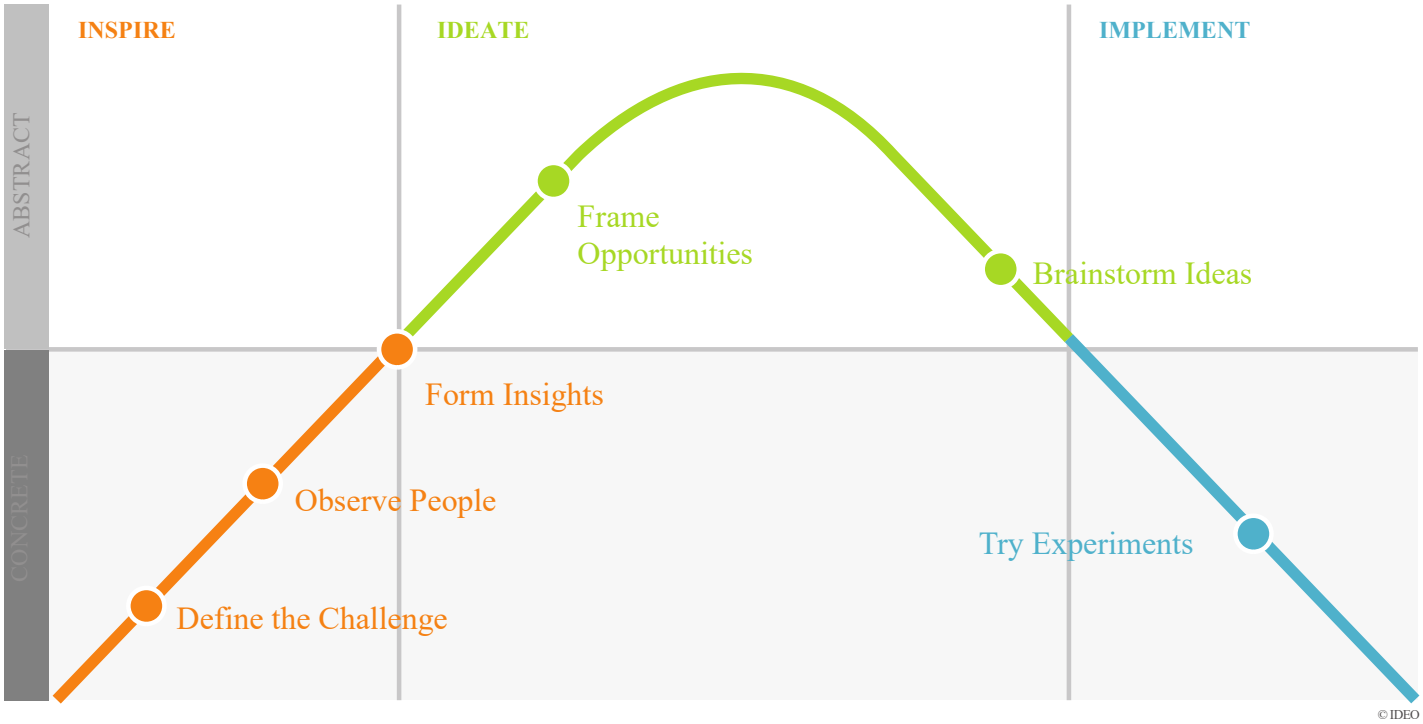
In this phase, multiple experiments may be necessary.

Once the idea shows some promise, additional design refinement is required along with testing the idea for its **viability** and **feasibility**.




1
DESIGN THE RIGHT THING
2
DESIGN THINGS RIGHT
PROBLEM
PROBLEM DEFINTION
SOLUTION
RESEARCH
DESIGN
DISCOVER
EXPLORE
DEFINE
DEVELOP
TEST
DELIVER
LISTEN
Insight into the Problem
Scope down the Focus
Potential Solutions
Solutions that Work & Receive Feedback

Design Thinking Innovation Process





OBSERVE PEOPLE

A lot of times people don’t know what they want until you show it to them.

— STEVE JOBS

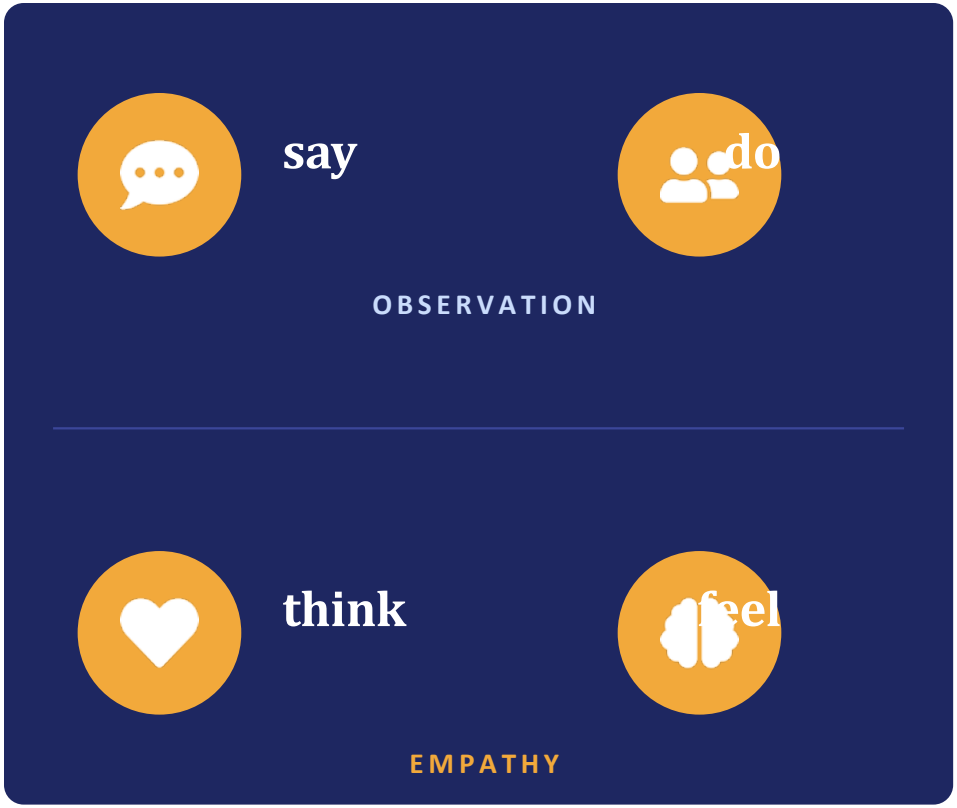


Observe people where they live, work and socialize for inspiration and to gain empathy.

What people say they do and what they actually do are often different.

Qualitative research enables you to put yourself in someone else's shoes to get beyond what they say to what they think and feel.

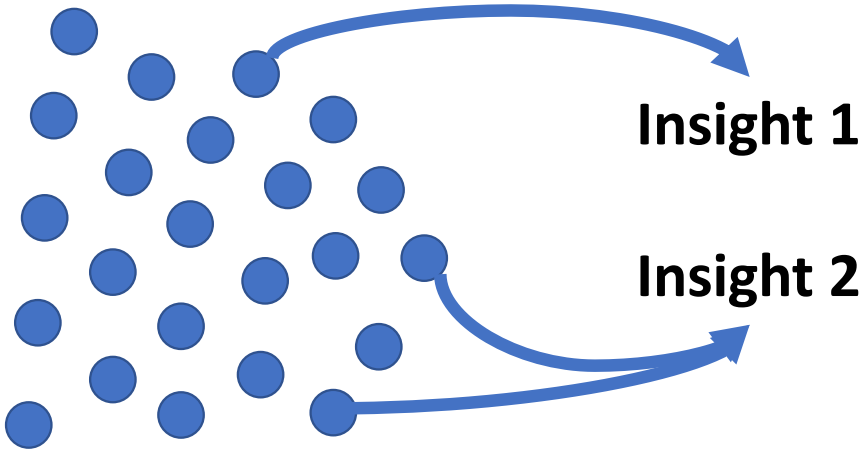
Today we will use mobile ethnography and extreme users as methods for observing people.



Insight Generation

OBSERVATIONS (FACTS)

INTERPRETATION (Why)





"Always learning something new in my Design Class. Whew!"



"Homemade white Pepper at work! Such a pleasure!"

INSIGHT

Millennials want their lives to be an exciting adventure, even if that excitement comes in small bits in their everyday lives

An Insight Is Not...

AN OBSERVATION

Observations are an incredibly important part of creating insights but are still only one data point to consider and should never stand alone.

Facts lack the "why" and the "motivation" behind a consumer's behaviour.

A USER'S WISH

Articulating user's needs is perfect for identifying new features and incremental changes, but do not result in insights which have the power to overwhelm existing solutions and generate new ones.

Insights are less apparent, intangible, latent.

A SOLUTION

Sometimes, there is a tendency to jump immediately to solutions or to formulate insights in a mission statements format.

The most powerful insights come from rigor and serious analysis to translate large amounts of data into concise and compelling findings.

How to Validate an Insight?



AUTHENTIC

Based on observations



NON OBVIOUS

Something that is latent and less apparent

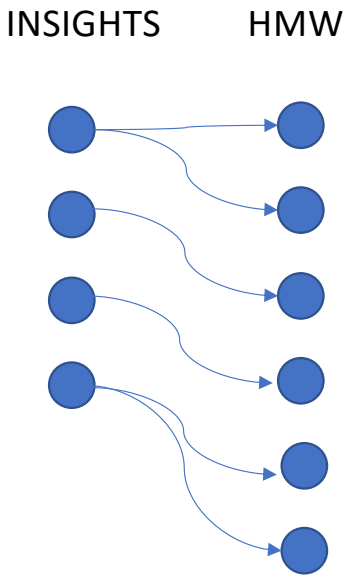


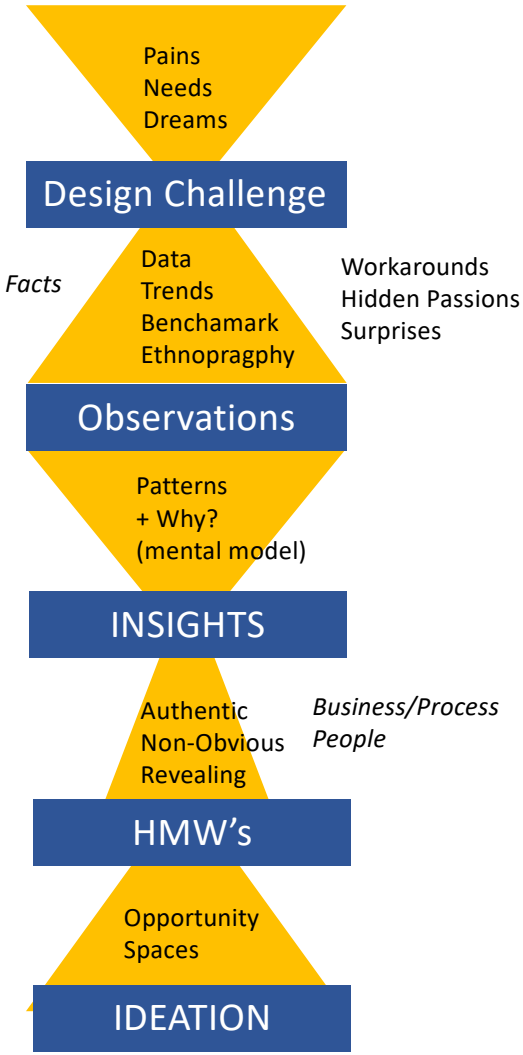
REVEALING

the “why” and the “motivation” behind a behaviour

How Might We

- Start by looking at the insight statements that you've created.
- What further insights/topics arise from the design challenge? These topics are subsets of the entire problem. They focus on different aspects of the challenge.
- Try rephrasing/reframing them as questions by adding "How might we" at the beginning.
- The goal is to find opportunities for design, so if your insights suggest several How Might We questions, that's great!





Select Your Best Idea Based on These Criteria



INSPIRATIONAL

Your idea should be novel, exciting, and easy to understand by someone not steeped in the process or the topic.



CONNECTED

Your idea should meet a real need and solve a real problem.



RELEVANT

Your idea should be relevant to the original design challenge.

Designing an Experiment

Before investing a lot of time and resources, design and run an experiment to test the desirability, viability, or feasibility of your idea.

LOW RISK

Conducted with a safe audience, at a minimal cost (guideline: under 100 euros).

QUICK & EASY

Easy to build, easy to run (guideline: under 1 hour).

GENERATIVE

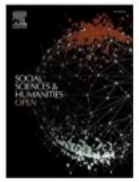
On target to answer a question that will grow the idea.

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Regular Article

Enhancing corporate innovation through generative AI: A case study on Human–AI collaboration

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ABSTRACT

Although Generative Artificial Intelligence (GenAI) is revolutionizing corporate innovation processes, little is known about how it fits into Design Thinking (DT). This study uses an exploratory case study in a corporate innovation workshop to examine how GenAI affects three essential DT components: trend identification, brainstorming, and feedback. Using a mixed-methods approach, we assessed participants' perceptions before and after DT activities with and without GenAI help. Our findings demonstrate that GenAI accelerates ideation, improves creative confidence, and promotes trend analysis and feedback integration. However, ethical concerns regarding AI bias and transparency also exist, which emphasizes how important it is to use AI ethically. Human judgement is necessary at every evaluation stage to ensure morally sound and contextually relevant decision-making, even while GenAI promotes efficiency and alternative thinking. This study contributes to the emerging discourse on AI-human collaboration in innovation, offering insights for organizations seeking to integrate GenAI into human-centered design processes.

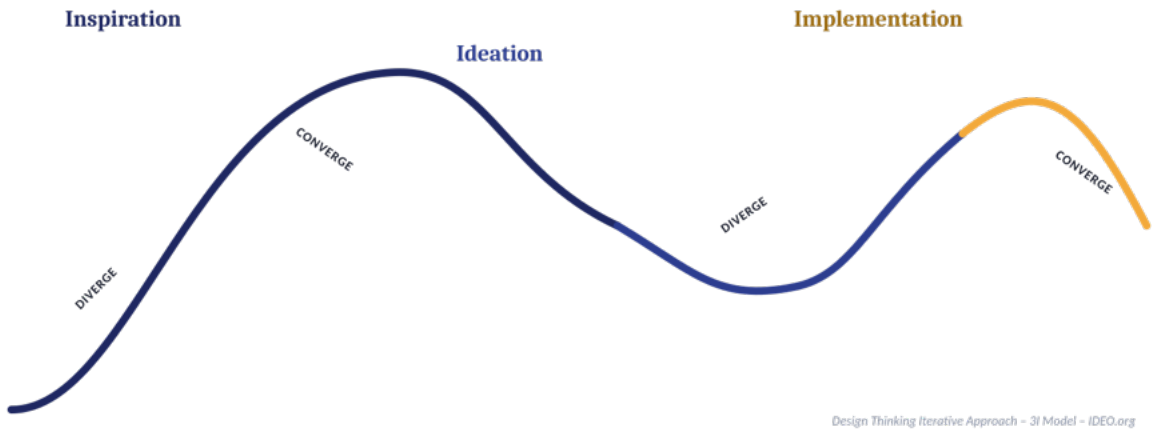
Purpose of the Study

AI Transformation in Industries

- AI revolutionizes efficiency and creativity across sectors *(Bughin et al., 2017).*
- Innovation is essential to corporate competitiveness and adapting to market demands.

Design Thinking as a Solution

- A flexible, human-centered process to address complex problems.
- Balances human needs, technology potential, and economic goals *(Brown, 2009).*



Purpose of the Study

Corporate Innovation Challenges

• Definition of innovation as creating, developing, and implementing value-driven concepts *(Chesbrough, 2003).*

• Innovation is vital for adapting to dynamic markets and meeting consumer needs *(Baregheh et al., 2009).*

• Key Challenge: Balancing Exploitation (current needs) vs. Exploration (future needs) for sustained success *(Corso & Pellegrini, 2007).*

Advances and Challenges of Corporate Innovation Studies

• Bridging theory and practice remains difficult, requiring new methodologies for quick adaptation *(Baregheh et al., 2009).*

• Emerging technologies like AI bring both opportunities and challenges to corporate innovation *(Fui-Hoon Nah et al., 2023).*

• AI's Role: AI can enhance empathy, creativity, and experience in innovation while aligning with core principles *(Verganti et al., 2020).*

The Innovator’s Dilemma

Corporate innovation means creating, developing and implementing value-driven concepts (Chesbrough, 2003) — and that constantly forces a choice.

EXPLOITATION

Refine what already works

Serve today’s customers better. Low risk, fast payoff — but it’s where yesterday’s winners get comfortable.



EXPLORATION

Take risks on what’s next

Anticipate needs that don’t exist yet. High uncertainty, long horizon — but it’s where tomorrow’s advantage comes from.

The dilemma: the approach that makes a company successful today is often exactly what makes it blind to the disruptive shift coming tomorrow (Corso & Pellegrini, 2007).

Notice the parallel: just like every Design Thinking phase must diverge before it can converge, a company has to keep exploring before it can safely exploit. Miss either half, and the dilemma wins.

Literature Review

Analysis of Challenges within the Design Thinking Process



Inspiration

Trend Spotting

ROLE

Identifying future consumer needs and market dynamics

IMPORTANCE

Reducing risks and optimizing strategies



Ideation

Brainstorming

ROLE

Generating ideas and fostering creativity

IMPORTANCE

Removing obstacles and encouraging innovation



Implementation

Feedback

ROLE

Supplying information and perfecting adjustments

IMPORTANCE

Ensuring project success and ongoing development

A design thinking process that incorporates trend identification, idea generation and feedback should be a comprehensive method to address difficulties and problems related to corporate innovation.

Literature Review

Through an in-depth exploration of the integration of generative AI in the design thinking, this study aims to:

Understand how generative AI affects the traditional stages of design thinking within corporate innovation

Understand the distinctive contribution that generative AI brings to the specific phases of the design thinking process

RESEARCH ASSUMPTIONS

1. Generative artificial intelligence enhances the overall design thinking process
2. The integration of generative artificial intelligence in the design thinking process enhances the output of the inspiration phase
3. The integration of generative artificial intelligence in the design thinking process enhances the output of the ideation phase
4. The integration of generative artificial intelligence in the design thinking process enhances the output of the implementation phase
5. The integration of generative artificial intelligence in the design thinking process increases individuals' creative confidence
6. The integration of generative artificial intelligence in the design thinking process increases openness to technological advancements
7. The integration of generative artificial intelligence in the design thinking process decreases ethical issues

Methodology Overview

A mixed methods approach with five steps that incorporates experimental procedures, quantitative and qualitative analysis, and pre and post test assessments to evaluate the role of generative artificial intelligence.

1

Pre-Test Questionnaire

Gather initial data and initial participants perceptions before the experiment

2

Experiment – Design Thinking Session

Design Thinking session with three exercises to assess the influence of artificial generative intelligence

3

Post-Test Questionnaire

Gather final data and final participants perceptions after the experiment.

4

Quantitative Analysis

Statistical analysis comparing pre-test and post-test results

5

Qualitative Analysis

Comparison between the in-session participants votes and their leadership votes

With this, we facilitated a robust integration of generative artificial intelligence into a human centered approach and its influence on participants creative confidence, openness to technological advancements and ethical considerations.

Methodology *(Pre and Post Test Questionnaires)*

Structure	Pre-Test Questionnaire	Post-Test Questionnaire
Purpose	Assess initial perceptions of generative AI's impact on the design thinking process.	Evaluate shifts in perceptions post-experiment.
Primary Information	Familiarity with design thinking and generative AI usage frequency.	—
Overall Perceptions	Views on generative AI in design thinking (e.g., trend identification, bias concerns, human vs. AI feedback, innovation).	*Same items, re-measured*
Design Thinking Phases	Inspiration: confidence in trend identification, openness to tech, ethical concerns, AI's usefulness in trend analysis. Ideation: confidence in idea generation, openness to tech, ethical concerns, AI's usefulness in brainstorming. Implementation: confidence in providing feedback, openness to tech, ethical concerns, AI in feedback processes.	*Same items, re-measured*
Demographics	Age, gender, education, and professional role to provide context for responses.	—
Response Format	Likert scale (1-7) for level of agreement.	*Same scale*

Methodology (Experiment)



Participants were divided into small groups, each with different specifications to test generative AI's role in the design thinking process:

"Humans"

could only use their own mental capacities to respond to the exercise.

"AI Enhanced"

they were obligated to use generative artificial intelligence but only after carrying out the exercise with their mental abilities.

"AI Free Access"

it was up to the group to decide whether to use generative artificial intelligence and how to utilize the resource.

"Non-expert with Chat GPT in back office"

conducted entirely in the back office by the researcher, where all inputs originated solely from Chat GPT.

Methodology (Experiment)

The session encompassed three key exercises corresponding to the different phases of design thinking:



INSPIRATION

1. Identifying Trends

Each small group was asked to identify trends that they believed would impact their challenge in the future, creating spaces of opportunity for value creation. After each small group discussion, each group submitted an identified opportunity space in a brief and simple sentence.



IDEATION

2. Brainstorming

In small groups, participants generated ideas that responded to the most voted opportunity space from the previous exercise. Each group then selected one idea they considered to have the greatest potential for implementation and value creation and submitted it in a brief sentence.



IMPLEMENTATION

3. Feedback

Each small group presented their final idea in a three minute pitch. The remaining groups provided feedback with one positive point and one point for improvement. Each group then selected the feedback received they found most useful.

Methodology (Experiment)



Voting Process

• After each 30-minute exercise, each small group submitted their results.

• Participants then voted individually on the best output, excluding their own group's submission. This voting approach reflects the "wisdom of the crowd" concept, where collective group decisions are often more accurate than those of individual experts *(Surowiecki, 2005).*

• In addition, leadership present in the session also voted on the best outputs from all groups despite not having carried out the exercises. Leaders voted without knowing which group produced each output, allowing for an unbiased comparison between participant (wisdom of the crowd) votes and leadership decisions.



Company Selection Rationale

Expertise in blending AI with human skills and CX analytics.

Emphasis on user-centric innovation aligns with design thinking principles.

Diverse participant selection from different departments fosters innovation through varied perspectives (Thoring et al., 2014).

Methodology (Quantitative Analysis)

Quantitative Analysis

The quantitative analysis aims to statistically assess changes in participants' perceptions of generative AI in design thinking—measuring creative confidence, openness to technology, ethical considerations, and overall views on AI integration before and after the experiment.



Methodology (Qualitative Analysis)



Qualitative Analysis

The qualitative analysis of this study aims to compare leadership votes with the wisdom of the crowd votes to assess how generative AI influences the design thinking process and fosters inclusive, innovation-driven decision-making in a corporate context.

Wisdom of the Crowd Choices

VS

Leadership Choices

Results

(Quantitative – Pre and Post Test Questionnaire Results)

Tests of Normality — passed

Phase	Variable		Results
Inspiration	Creative Confidence	▲	**Significant increase**
	Openness to Technological Advancements	—	No change, but initial level high
	Ethical Considerations & Overall Perceptions	—	Stable, no significant changes
Ideation	Creative Confidence	▲	**Significant increase**
	Openness to Technological Advancements	—	No change, initial level high
	Ethical Considerations	—	Stable, no significant changes
Implementation	Openness to Technological Advancements	▲	**Significant increase**
	Creative Confidence & Overall Perceptions	—	Stable, slight increase but not significant
	Ethical Considerations	—	No significant changes
Overall Perceptions of DT with AI Integration	AI for Identifying Trends	▲	**Significant increase**
	AI Influencing Solutions, Providing Feedback, Enhancing Innovation	—	Stable, initial perceptions high

Results

(Quantitative – Pre and Post Test Questionnaire Results)

Correlation Analysis

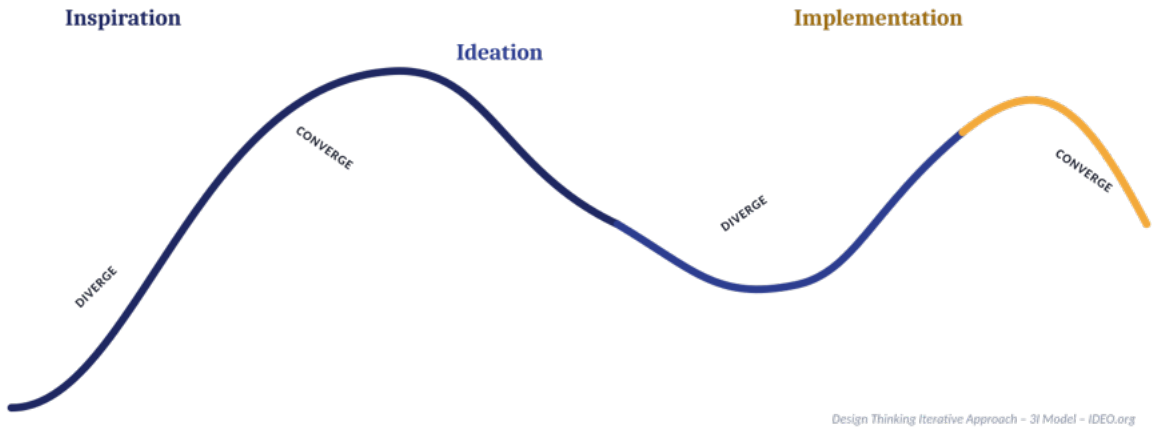
No statistically significant correlations were found between the belief that generative AI enhances innovation and:

- Familiarity with Design Thinking
- Belief in Design Thinking's suitability for addressing corporate innovation problems
- Regularity of using generative AI
- Familiarity with generative AI

Results

(Qualitative – Experiment: Design Thinking Session)

The experiment insights taken from the results are the following:



"AI Free Access" group won in both crowd and leadership evaluations.

Suggests that unrestricted AI use enhances creativity and divergent thinking.

"Non Expert with ChatGPT alone in Back Office" group won in crowd evaluation, while "AI Free Access" was preferred by leadership.

Indicates that AI can independently generate viable ideas and, supports diverse and innovative ideation.

Mixed results across feedback groups

Highlights the effectiveness of integrating AI feedback with human insights to refine and implement ideas.

Discussion

Inspiration and Ideation Phases

Generative AI significantly boosts creative confidence, supporting higher rates of innovation and idea generation. Participants feel more capable and empowered with AI assistance, enhancing creativity in these phases.

Implementation Phase

AI integration increases openness to technological advancements when participants see tangible benefits and receive useful feedback. Effective integration requires a balance of human insights and AI capabilities, fostering innovation.

Human-AI Collaboration

Combining generative AI with human expertise yields superior results across various disciplines. Future success in professional fields depends on merging human creativity with AI's computational strengths.

Initial Positive Views on AI

Observed changes post-experiment are limited by participants' already high regard for AI's role in design thinking.

Conclusions

RESEARCH QUESTION

How does the integration of generative artificial intelligence affect the traditional stages of the design thinking process, including inspiration, ideation, and implementation, within corporate environments?

Generative AI greatly improves the traditional phases of the design thinking process. It increases creative confidence during the inspiration phase, enhances idea generation during the ideation phase, and improves receptivity to technological advancements during the implementation phase. Generative AI boosts creative confidence, helps in trend identification, encourages idea generation, and improves feedback receptivity, although it does not significantly affect ethical considerations.

RESEARCH QUESTION

What distinctive contribution does generative artificial intelligence bring to the specific stages of a design thinking process?

In the inspiration phase, generative AI helps identify future needs; in the ideation phase, it stimulates a broad range of novel ideas; and in the implementation phase, it facilitates the exchange of feedback. There are the unique contributions of generative AI to each phase. At every stage of the design thinking process, generative AI provides distinctive advantages that promote innovation.

Conclusions

#	Research Assumptions	Conclusions
1	Generative artificial intelligence enhances the overall design thinking process	This was partially confirmed. Generative AI significantly increased creative confidence in the inspiration and ideation phases.
2	The integration of generative artificial intelligence in the design thinking process enhances the output of the inspiration phase	Confirmed. Participants showed a significant increase in creative confidence during the inspiration phase.
3	The integration of generative artificial intelligence in the design thinking process enhances the output of the ideation phase	Confirmed. There was a significant improvement in the generation of novel ideas.
4	The integration of generative artificial intelligence in the design thinking process enhances the output of the implementation phase	Partially confirmed. While there was increased openess to technological advancements, the impact on ethical considerations and the overall perception of generative AI feedback was not significant.
5	The integration of generative artificial intelligence in the design thinking process increases individuals' creative confidence	Confirmed in the inspiration and ideation phases but not significantly in the implementation phase.
6	The integration of generative artificial intelligence in the design thinking process increases openness to technological advancements	Partially confirmed. Confirmed in the implementation phase.
7	The integration of generative artificial intelligence in the design thinking process decreases ethical issues	Not confirmed. There was no significant change in ethical considerations.

- Integrating generative AI as a tool in the design thinking process is an improvement, especially during the inspiration and ideation phases.
- Generative AI significantly boosts creative confidence and facilitates the generation of novel ideas, while having little effect on ethical issues.
- Generative AI successfully encourages receptivity to technological advancements when they are being implemented.
- The collaboration between humans and generative AI shows how important is to integrate generative AI to foster an innovative culture in corporate settings.

Future Research

Study Limitations

- Sample size
- Single Company Data
- Selection Bias
- Self-Reported Data
- Qualitative Analysis: Lacks detailed methodology for comparing leader and group decisions, affecting rigor (Yin, 2015).
- Chat GPT Prompts: Created by principal investigator, leading to variability in prompt creation, which may affect consistency.

Future Research Directions

- Investigate AI's integration across various industries for specialized insights.
- Conduct longitudinal studies to observe AI's influence on design thinking over time.
- Explore ethical implications and develop strategies to address them.
- Use larger, more diverse samples to enhance universality of findings.